EE561 Advanced Static Power Conversion

Multilevel Converters: Modulation and Balancing — Quiz Answer Key

For the presenter and the instructor · not for distribution to students

Question 1 · DC-bus utilization and min-max injection — 25 pts

(a) $V_{1,pk}$ = 380 · √2 / √3 = 310.3 V; m = 310.3 / 300 = **1.034**.

(b) Sine PWM: **no** — m > 1, the references exceed the carriers (overmodulation, low-order harmonics). Min-max: **yes** — 1.034 < 2/√3 = 1.155.

(c) max = 0.98, min = −0.73 → r_0 = −(0.98 − 0.73)/2 = **−0.125**; r' = (0.855, −0.375, −0.855). $r'_a - r'_b = 1.23 = r_a - r_b$: the common offset cancels in every line-to-line difference.

(d) $V_{LL,max}$ = √3 · (2/√3)(V_{dc}/2) / √2 = V_{dc}/√2 = **424 V** (sine PWM: 367 V).

Marking: (a) 6; (b) 2.5 + 2.5; (c) r_0 3, r' 3, check 3; (d) 5.

Question 2 · Neutral-point balancing — 25 pts

(a) $\langle i_O \rangle = \Sigma(1 - |r_x + r_0|)\, i_x$. For a small offset that does not change the sign of r_x: $|r_x + r_0| = |r_x| + \text{sign}(r_x)\, r_0$, so $\Delta\langle i_O \rangle = -r_0\, \Sigma\, \text{sign}(r_x)\, i_x$.

(b) Averaged over a period, $\Sigma\, \text{sign}(r_x)\, i_x = (6/\pi)\, \hat{I} \cos\phi$. At $\phi = 90°$ this is ≈ 0: the offset no longer changes the average midpoint current, so **ΔV is not restored** (only the $3f_1$ ripple remains). Low power factor needs another method (redundant-vector selection, larger capacitors).

(c) The sensitivity $-(6/\pi)\, \hat{I} \cos\phi$ **changes sign with the power direction**. In rectifier mode ($\cos\phi < 0$) the loop tuned as negative feedback becomes **positive feedback**: ΔV runs away until the overvoltage protection trips. Fix: $r_0 = \text{sign}(P) \cdot K \cdot \Delta V$ (or a current-direction-aware state selection), and test both power directions.

Marking: (a) 8; (b) prediction 3, explanation via cos ϕ 5; (c) sign change 4, consequence 2, fix 3.